

## BLOCK ISLAND STATE, RI, USA

WMO: 725058

Lat: 41.168N

Lon: 71.578W

Elev: 105

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94793

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.7       | 17.0 | -6.4                          | 3.9 | 15.1 | -0.8 | 5.3 | 19.3 | 31.5                  | 37.7 | 28.3 | 36.7 | 14.5                  | 330  | 0.595 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.4                   | 82.1            | 72.4 | 80.7 | 71.5 | 78.6 | 70.4 | 75.2                | 79.8 | 73.9 | 77.8 | 72.6 | 76.2 | 11.1                 | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 73.4                            | 124.6 | 77.2 | 72.5 | 120.9 | 76.4 | 71.9 | 118.2 | 75.7 | 38.6          | 80.1 | 37.5 | 78.3 | 36.3 | 76.0 | 80.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 26.0              | 23.4 | 20.6 | 6.6                        | 86.6 | 4.2                | 3.3 | 3.6                                                | 89.0 | 1.2        | 91.0 | -1.1       | 92.9 | -4.2       | 95.3 |
|                   |      |      | 4.6                        | 76.4 | 4.3                | 2.0 | 1.6                                                | 77.8 | -0.9       | 79.0 | -3.3       | 80.1 | -6.4       | 81.5 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 52.0   | 32.6  | 33.7  | 38.4  | 46.8  | 55.4  | 64.8  | 71.8  | 71.2  | 65.8  | 56.3  | 46.9  | 39.1 |
|                                                                                  | DBStd       | 15.12  | 9.46  | 7.93  | 6.76  | 5.32  | 5.41  | 5.31  | 3.89  | 4.09  | 4.80  | 6.32  | 7.37  | 7.58 |
|                                                                                  | HDD50       | 1993   | 540   | 456   | 361   | 121   | 10    | 0     | 0     | 0     | 0     | 18    | 142   | 345  |
|                                                                                  | HDD65       | 5293   | 1003  | 876   | 823   | 545   | 303   | 67    | 2     | 3     | 49    | 276   | 544   | 802  |
|                                                                                  | CDD50       | 2727   | 1     | 0     | 3     | 26    | 176   | 445   | 677   | 656   | 473   | 213   | 49    | 8    |
|                                                                                  | CDD65       | 551    | 0     | 0     | 0     | 0     | 4     | 62    | 214   | 194   | 71    | 6     | 0     | 0    |
|                                                                                  | CDH74       | 1746   | 0     | 0     | 0     | 2     | 12    | 160   | 802   | 666   | 102   | 2     | 0     | 0    |
|                                                                                  | CDH80       | 213    | 0     | 0     | 0     | 0     | 0     | 18    | 115   | 75    | 5     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 11.4   | 13.0  | 12.6  | 12.5  | 11.8  | 11.1  | 10.5  | 10.0  | 9.3   | 10.0  | 11.4  | 12.3  | 12.5 |
| Precipitation                                                                    | PrecAvg     | 40.3   | 3.3   | 3.2   | 3.7   | 3.6   | 3.3   | 2.8   | 2.7   | 3.2   | 3.5   | 2.9   | 4.1   | 3.9  |
|                                                                                  | PrecMax     | 59.6   | 8.8   | 6.9   | 8.0   | 9.2   | 6.1   | 8.7   | 7.1   | 9.0   | 11.5  | 7.3   | 9.1   | 8.1  |
|                                                                                  | PrecMin     | 24.1   | 0.2   | 0.8   | 1.1   | 0.8   | 1.4   | 0.6   | 0.3   | 0.2   | 0.3   | 0.8   | 0.9   | 0.9  |
|                                                                                  | PrecStd     | 7.6    | 2.0   | 1.6   | 1.8   | 2.0   | 1.6   | 2.1   | 1.7   | 2.1   | 2.5   | 1.4   | 2.3   | 1.9  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 55.2  | 52.5  | 58.6  | 65.1  | 75.1  | 82.1  | 87.6  | 85.7  | 80.6  | 72.7  | 64.0  | 59.0 |
|                                                                                  |             | MCWB   | 53.3  | 49.9  | 50.7  | 55.6  | 63.4  | 70.6  | 75.2  | 74.6  | 71.3  | 66.3  | 60.4  | 56.1 |
|                                                                                  | 2%          | DB     | 52.4  | 49.3  | 53.8  | 60.6  | 70.5  | 78.8  | 82.3  | 81.8  | 76.8  | 70.0  | 62.3  | 55.4 |
|                                                                                  |             | MCWB   | 50.8  | 46.4  | 47.9  | 51.4  | 61.7  | 68.3  | 72.6  | 72.6  | 69.7  | 65.3  | 58.7  | 53.0 |
|                                                                                  | 5%          | DB     | 48.3  | 46.5  | 50.1  | 56.7  | 66.0  | 74.9  | 81.0  | 79.9  | 74.5  | 67.4  | 60.6  | 54.2 |
|                                                                                  |             | MCWB   | 46.4  | 43.5  | 46.0  | 49.6  | 59.1  | 66.6  | 71.9  | 71.4  | 69.0  | 64.0  | 56.9  | 52.0 |
|                                                                                  | 10%         | DB     | 45.7  | 44.9  | 47.6  | 54.5  | 63.5  | 72.6  | 78.9  | 78.3  | 72.7  | 64.9  | 57.3  | 51.8 |
|                                                                                  |             | MCWB   | 42.8  | 41.2  | 44.3  | 48.4  | 57.2  | 65.5  | 70.8  | 70.6  | 68.0  | 61.2  | 53.7  | 49.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 54.1  | 51.3  | 52.9  | 57.8  | 66.4  | 73.0  | 76.8  | 77.2  | 74.5  | 68.6  | 62.5  | 57.2 |
|                                                                                  |             | MCDB   | 55.0  | 51.8  | 55.4  | 62.7  | 71.7  | 78.5  | 83.3  | 82.3  | 78.4  | 70.6  | 63.5  | 58.6 |
|                                                                                  | 2%          | WB     | 51.0  | 47.2  | 49.5  | 54.4  | 63.4  | 70.6  | 75.2  | 75.4  | 72.5  | 66.5  | 59.5  | 54.3 |
|                                                                                  |             | MCDB   | 52.3  | 48.2  | 52.5  | 57.5  | 67.9  | 75.7  | 80.2  | 79.5  | 75.0  | 68.5  | 61.7  | 55.5 |
|                                                                                  | 5%          | WB     | 46.6  | 44.9  | 46.5  | 52.1  | 60.8  | 68.8  | 74.1  | 74.1  | 70.9  | 64.4  | 57.4  | 52.0 |
|                                                                                  |             | MCDB   | 47.7  | 46.4  | 48.5  | 55.0  | 65.2  | 73.1  | 78.4  | 77.7  | 73.2  | 66.1  | 59.7  | 54.0 |
|                                                                                  | 10%         | WB     | 43.5  | 41.3  | 44.4  | 49.7  | 58.5  | 66.8  | 72.8  | 72.6  | 69.3  | 62.4  | 55.1  | 48.6 |
|                                                                                  |             | MCDB   | 45.6  | 44.3  | 46.9  | 53.4  | 62.7  | 70.9  | 76.7  | 76.1  | 71.7  | 64.6  | 57.7  | 50.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.0  | 10.7  | 10.7  | 11.1  | 11.3  | 11.3  | 11.4  | 10.8  | 10.0  | 10.2  | 10.3  | 10.2 |
|                                                                                  |             | MCDBR  | 13.8  | 12.6  | 13.4  | 14.1  | 15.9  | 14.9  | 13.7  | 12.6  | 11.4  | 11.2  | 10.7  | 12.1 |
|                                                                                  |             | MCWBR  | 14.5  | 12.0  | 10.6  | 9.1   | 9.4   | 8.5   | 7.2   | 6.8   | 7.7   | 9.2   | 11.0  | 13.4 |
|                                                                                  | 5% WB       | MCDBR  | 13.6  | 13.1  | 12.8  | 12.5  | 13.6  | 12.8  | 11.6  | 10.4  | 9.8   | 10.1  | 10.3  | 12.7 |
|                                                                                  |             | MCWBR  | 14.6  | 13.8  | 11.9  | 10.0  | 9.3   | 8.3   | 6.9   | 6.7   | 7.7   | 10.1  | 11.3  | 14.5 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.313  | 0.321 | 0.343 | 0.377 | 0.424 | 0.463 | 0.488 | 0.449 | 0.381 | 0.352 | 0.321 | 0.310 |      |
|                                                                                  | taud        | 2.497  | 2.473 | 2.448 | 2.388 | 2.290 | 2.223 | 2.160 | 2.266 | 2.428 | 2.509 | 2.550 | 2.549 |      |
|                                                                                  | Ebn at Noon | 268    | 283   | 288   | 284   | 272   | 260   | 252   | 259   | 272   | 268   | 263   | 260   |      |
|                                                                                  | Edh at Noon | 24     | 28    | 32    | 36    | 41    | 44    | 47    | 41    | 33    | 27    | 23    | 21    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 582    | 870   | 1210  | 1567  | 1738  | 1904  | 1939  | 1746  | 1422  | 965   | 649   | 491   |      |
|                                                                                  | RadStd      | 48     | 87    | 93    | 167   | 168   | 188   | 105   | 100   | 108   | 88    | 49    | 46    |      |

## Historical Trends

|              |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |
|--------------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|
|              |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |
|              |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |
| Station Only |                         | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | N/A |
|              | Regional (64 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | +180  | +95 |

Nomenclature: See separate page